

UThread: A User-Space Thread Scheduling Engine

High-Performance M:N Green Thread Runtime with Work Stealing, Async I/O & Preemptive Scheduling in C

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PROJECT OVERVIEW

UThread is a production-grade, user-space thread scheduling library written entirely in C. Modern high-concurrency systems — such as Go's Goroutine scheduler and Java's Project Loom — rely on green threads and M:N threading models to achieve massive scalability far beyond what the OS kernel can provide natively. UThread was designed and built from scratch to replicate, explore, and benchmark these real-world concurrency mechanisms. The library manages thread context switching entirely in user-space using the POSIX `ucontext_t` API, achieving **sub-microsecond context switch latencies** — approximately **7.75x faster** than native Linux pthreads — and can spawn and schedule **15,000+ concurrent threads** in a fraction of a second. The project evolved through 14 development phases, from a basic cooperative 1:N scheduler into a fully featured M:N work-stealing engine with preemptive scheduling, synchronization primitives, asynchronous I/O, memory-safe guard pages, and interactive execution tracing.

CORE FEATURES

M:N Work Stealing

Maps thousands of user threads across multiple Virtual Processors (CPU cores). Idle cores proactively steal work from overloaded cores for automatic load balancing.

Preemptive Scheduling

SIGALRM-based timer interrupts force context switches every 10ms. Supports pluggable Round-Robin and Priority scheduling policies.

Synchronization Primitives

Mutexes, Counting Semaphores, and Buffered Channels built on scheduler-level blocking — no busy-waiting or spinlocks for user code.

Async I/O (epoll)

Non-blocking sleep, read & write APIs via Linux epoll. I/O-bound threads yield automatically; the CPU never sits idle.

Guard Page Protection

Stacks allocated via mmap with mprotect guard pages. Custom SIGSEGV handler safely detects and reports stack overflows.

Execution Tracing

Ring buffer records all context switches, steals & I/O events. Exports Chrome Trace JSON for interactive Gantt chart visualization.

TECHNOLOGIES & TOOLS

C11

ucontext.h

pthreads

epoll

mmap / mprotect

SIGALRM

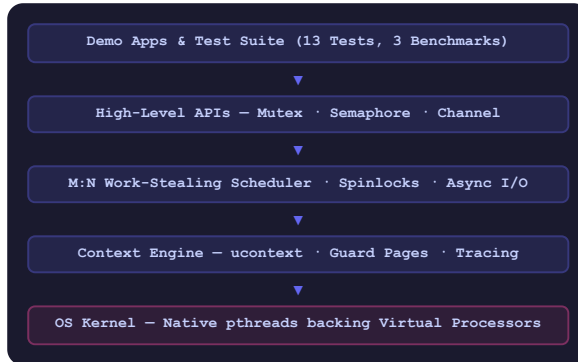
Atomic Ops

GCC

Make

Linux / WSL

ARCHITECTURE



BENCHMARK RESULTS

BENCHMARK	UTHREAD	PTHREADS	SPEEDUP
Context Switch (100K)	0.59s	4.58s	7.75x
Spawn 1K Threads	0.015s	0.241s	16.35x
M:N 4-VP vs 1:N	0.067s	0.147s	2.20x
15K Concurrent Threads	< 0.1s	Several sec	~15x

Context switch latency: ~1.85 µs/switch. Benchmarks on Linux x86-64.

KEY DESIGN DECISIONS

ucontext vs Assembly: We chose POSIX `ucontext_t` for portable, hardware-agnostic context switching over inline assembly, trading a small performance overhead for full cross-platform safety.

Scheduler-Level Blocking: Mutexes use scheduler-managed state transitions (RUNNING → BLOCKED → READY) via `swapcontext` instead of Peterson's-style busy-waiting, which we proved fails on modern CPUs due to memory reordering.

Dynamic TCB Allocation: Threads are dynamically allocated via intrusive linked-lists with O(1) garbage collection, removing all hardcoded limits and enabling infinite thread creation.

CODEBASE STATISTICS

~1,200 lines of core scheduler code (uthread.c) | 150 lines of public API header | 13 test programs | 3 benchmark suites | 1 interactive Gantt Chart visualization tool